

Abstract No.

Repurposing statins as senolytics in pediatric low-grade glioma leverages mevalonate pathway dependency in oncogene-induced senescence

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Pediatric low-grade glioma (pLGG) is driven by genetic alterations of the MAPK pathway, that initiate oncogene-induced senescence (OIS) in the majority of pLGG tumor cells. Current therapy, including chemotherapy and MAPK inhibitors, does not target tumor cells in OIS, which serve as source for relapse after therapy cessation. Hence, eradication of senescent tumor cells has the potential to induce durable responses. Our group has shown that BCL-XL is a targetable vulnerability of senescent pLGG cells, but so far clinical translation has been hampered by the lack of approved BCL-XL inhibitors. To identify novel translationally relevant dependencies, we performed a drug repurposing screen using a library comprising approved and brain-penetrant drugs. n=564 drugs were screened in three patient-derived pLGG cell lines. Using the iTreX platform, drug sensitivity scores were calculated to compare drug response between OIS and proliferation. Top hit drugs were validated using metabolic activity assays and high-content microscopy. Drug interaction with standard of care therapy and BCL-XL inhibitors was assessed. Analysis of gene and protein expression and pathway rescue experiments were used to investigate downstream metabolic dependencies and the mode of cell death induction. The two HMG-CoA-reductase inhibitors (statins) lovastatin and atorvastatin, that block the mevalonate pathway, were identified as top hit drugs with senolytic activity. Intriguingly, in-class senolytic properties were validated with brain-penetrant and pediatric approved statins, fluvastatin and pitavastatin, at clinically achievable concentrations. Synergy assessment revealed enhanced senolysis in combination with BCL-XL inhibitors. Rescue of statin-induced cytotoxicity with mevalonate verified on-target activity. Among downstream mevalonate pathway metabolites, geranylgeranyl pyrophosphate (GGPP) was identified as a survival dependency. In summary, statins were identified as a novel and clinically approved senolytic drug class in pLGG with high translational potential. The mevalonate-GGPP pathway, a critical regulator of autophagy and vesicle trafficking, was discovered as an actionable pathway in senescent pLGG.



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